

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	g	t	a	b	
s2	g	x	t	x	a	y	b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main()
5  {
6      char s1[100], s2[100];
7      int dp[100][100];
8
9      scanf("%s", s1);
10     scanf("%s", s2);
11
12     int n = strlen(s1);
13     int m = strlen(s2);
14
15     for (int i = 0; i <= n; i++)
16     {
17         for (int j = 0; j <= m; j++)
18         {
19             if (i == 0 || j == 0)
20                 dp[i][j] = 0;
21             else if (s1[i - 1] == s2[j - 1])
22                 dp[i][j] = dp[i - 1][j - 1] + 1;
23             else
24             {
25                 if (dp[i - 1][j] > dp[i][j - 1])
26                     dp[i][j] = dp[i - 1][j];
27                 else
28                     dp[i][j] = dp[i][j - 1];
29             }
30         }
31     }
32
33     printf("%d", dp[n][m]);
34
35     return 0;
36 }
37
```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓